A Mechanistic Study of AI-Detection Features in Frozen BERT: Sparse Probing and Activation Patching on RAID

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Abstract

[TODO: Write abstract last, after Section 1 is stable.]

1 Introduction

The rapid development of Large Language Models (LLMs) has made AI-generated text increasingly difficult to distinguish from human writing (Dugan et al., 2024). This poses a significant challenge across domains ranging from academic integrity to social media moderation and journalist authenticity. Even though the detection accuracy has improved substantially, with encoder-based classifiers reaching high performance on established benchmarks (Dugan et al., 2024), a fundamental question remains unanswered: *What internal representations actually drive these predictions?* Mechanistic interpretability has seen little application to AI-text detection, leaving the field without a causal account of how detection actually works.

Existing interpretability work on AI-text detection takes one of three forms: a) *black-box feature attribution*, which identifies input-level correlates via SHAP or LIME without inspecting model internals (Pudasaini et al., 2026); b) *confounder removal*, which suppresses neurons correlated with domain or style artifacts in fine-tuned encoders to improve out-of-distribution accuracy (Borile and Abrate, 2025); or (c) *sparse autoencoder feature analysis* applied to decoder-based models (Kuznetsov et al., 2025). These approaches share a fundamental limitation: they are *observational*. They identify neurons that correlate with detector behavior or remove features that hurt generalization, but none directly tests whether a small set of internal representations is *causally sufficient* to produce AI-text predictions.

We address this gap with a causal intervention study on a *frozen* BERT-base-uncased. Rather than fine-tuning the model for detection, we train sparse linear probes on pretrained CLS-token activations concatenated across all 12 layers (9,216 features total), following the sparse-probing protocol of Gurnee et al. (2023). An L1-regularized logistic regression selects a small stable set of candidate neurons; an independent L2 probe serves as the frozen evaluator for all subsequent experiments. We then apply two causal interventions to the same selected feature set: *mean ablation* which removes these features to test whether they are necessary, and *same-domain donor* $\rightarrow$ *target activation patching* replacing their activations from AI into human samples to test whether they are *sufficient* to flip detector predictions. The patching direction is the key distinction from prior work: rather than removing suspected confounders, we inject candidate detection signal and measure the result.

We find that it is sufficient. Across five generators spanning one base and four chat-RLHF model families, patching $\sim$50 selected neurons flips human $\rightarrow$ AI predictions at 1.07–8.15%, 10–16$\times$ above a random-feature baseline. This finding provides the first causal-sufficiency evidence for AI-detection circuits in a frozen pretrained encoder. Mean-ablation of the same features causes only 0.03–1.06 pp accuracy loss, revealing that the detection signal is *redundantly distributed*: easy to inject, hard to erase. Cross-generator analysis further shows that stable feature sets are largely generator-specific (pairwise Jaccard 0.00–0.24) yet cluster $\sim$32$\times$ above chance, with chat-RLHF generators concentrating in layer 12 while the sole base/non-RLHF generator (GPT-2) peaks at layer 11.

Our work is most directly related to Borile and Abrate (2025), who suppress confounding neurons in a fine-tuned encoder to improve out-of-distribution detection, an observational approach that tests necessity-for-confounding rather than causal sufficiency. We discuss the relationship to this and other prior work in Section 2.

Contributions.

1. A sparse-probing characterization of AI-detection signal in frozen BERT-base-uncased across five generators on RAID, identifying stable sets of 45-62 neurons ($<$1% of 9,216 features) sufficient for 90.5-98.8% detection accuracy with high cross-fold stability (Jaccard 0.67-0.76), with hyperparameters validated via a multi-generator stability sweep.

2. The first causal-sufficiency test for AI-detection features via same-domain donor $\rightarrow$ target activation patching, yielding flip rates 10-16$\times$ above a random-feature baseline across all five generators.

3. A cross-generator circuit analysis revealing generator-specific stable sets that nonetheless overlap $\sim$32$\times$ above chance, with a chat-RLHF (layer 12) vs. base/non-RLHF (layer 11) layer asymmetry.

4. Leave-one-family-out (LOGO) generalization showing that $\leq$1% of neurons carries 87-94% of full-feature detection signal across five unseen model families.

2 Related Work

2.1 AI-Generated Text Detection

Supervised detectors fine-tune pretrained language models as binary classifiers on pairs of human and machine-generated text. The GPT-2 output detector (Solaiman et al., 2019) established this paradigm by fine-tuning RoBERTa on GPT-2 samples; subsequent work extended the approach to instruction-tuned and RLHF-aligned generators with larger and more diverse training sets. These classifiers achieve high in-distribution accuracy but degrade substantially on generators or domains unseen during training (Dugan et al., 2024).

Zero-shot methods exploit statistical properties of language model distributions without requiring labelled data. DetectGPT (Mitchell et al., 2023) tests whether candidate text lies at a local maximum of log-probability relative to nearby perturbations, reflecting the tendency of model-generated text to occupy high-probability regions. Binoculars (Hans et al., 2024) uses the cross-perplexity ratio between two models as a calibrated detection score, achieving strong performance across generators. Both approaches require whitebox or API access to a reference model.

Systematic benchmarking has revealed that no single method generalises reliably. RAID dataset (Dugan et al., 2024) spans 11 generators and 6 domains with adversarial rephrasing attacks; classifiers that perform well in-distribution consistently fail on held-out generator families or domains. This pattern motivates our focus: rather than improving detection accuracy, we ask what internal representations actually drive it.

2.2 Interpretability of Text Detectors

The shallowest form of interpretability for AI-text detectors uses input-level attribution. Pudasaini et al. (2026) apply SHAP to explain classifier decisions, finding that the most predictive tokens diverge substantially across domains. This approach characterises the model's decision surface but cannot speak to what internal features underlie it.

More recent work examines representations inside the encoder. Borile and Abrate (2025) identify neurons in a fine-tuned BERT encoder whose activations correlate with domain or style artefacts rather than AI-authorship signal, and suppress them to improve out-of-distribution accuracy. Kuznetsov et al. (2024) take a geometric approach, removing a learned embedding subspace to produce a detector more robust to lexical variation. Both works operate on fine-tuned models and test necessity-for-confounding: the question is which features hurt generalisation, not which features are sufficient to drive predictions. Our work differs from Borile and Abrate (2025) on four axes: we use a *frozen* rather than fine-tuned encoder; we probe full-layer concatenated CLS representations rather than intermediate FFN activations; we apply activation patching to test *sufficiency* rather than ablation to identify confounders; and we evaluate on RAID rather than DAIGT/M4/HC3.

Kuznetsov et al. (2025) apply sparse autoencoders to Gemma-2 decoder activations, extracting semantically interpretable features correlated with AI authorship. This work is complementary but distinct: the model class (autoregressive decoder), representation type (token-position activations rather than CLS pooling), and analysis method (SAE reconstruction) all differ from ours, and the approach remains observational – features are identified by correlation, not by causal intervention. None of the above provides direct evidence that a small identified set of neurons is *sufficient* to produce AI-text predictions when injected into human-text activations; that is the question we address.

2.3 Sparse Probing and Activation Patching

Sparse probing uses L1-regularised linear classifiers to identify a small set of neurons in a pretrained model that carry task-relevant information. Gurnee et al. (2023) introduced and systematically evaluated the L1-then-L2 protocol: an L1 probe selects candidate neurons, and a re-fitted L2 probe on those neurons provides unbiased accuracy estimates. Their study demonstrates that individual neurons in GPT-2 and LLaMA encode grammatical, factual, and linguistic properties, and that the selected sets are stable across seeds and data subsets. We adopt this protocol directly, applying it to BERT-base-uncased CLS-token activations concatenated across all 12 layers.

Activation patching replaces internal representations in a target forward pass with those from a donor pass, then measures the effect on model outputs. Vig et al. (2020) introduced causal mediation analysis for language models, using interventions on attention heads to trace gender bias; Meng et al. (2022) applied patching at MLP layers in GPT to localise factual associations. Heimersheim and Nanda (2024) provide a systematic treatment of patching variants and recommend donor patching – replacing with activations from a real input rather than a zero or mean vector – to avoid out-of-distribution artefacts, and mean ablation as a conservative necessity baseline. We follow both recommendations: selected neurons are mean-ablated for the necessity test and replaced with activations from same-domain AI donors for the sufficiency test.

The closest methodological precedent for our flip-rate framing is Ravindran (2025), who apply donor-style activation patching to test whether deceptive activations from an aligned model can cause unsafe outputs in a target pass. The setting differs substantially – autoregressive decoder, safety alignment rather than AI-text detection – but the core intervention and the use of prediction-flip rate as the primary metric are shared. To our knowledge, no prior work applies this methodology to encoder-based AI-text detection.

3 Setup

3.1 Data: RAID

RAID (Dugan et al., 2024) provides human-written and machine-generated texts across 11 generators and 8 domains. We use six of the eight domains: abstracts, books, news, Reddit posts, reviews, and Wikipedia. We select six generators for architectural diversity: GPT-4 (`gpt4`), GPT-2 (`gpt2`), MPT-7B (`mpt`), Mistral-7B-Instruct (`mistral-chat`), Llama-2-Chat (`llama-chat`), and Cohere Command (`cohere-chat`), spanning two pure-base generators (gpt2, mpt), one SFT-only generator (mistral-chat), and three SFT+RLHF/preference-trained generators (gpt4, llama-chat, cohere-chat), across five model families. We subsample 7,500 examples per generator (3,750 human, 3,750 AI) balanced across the six RAID domains, yielding 45,000 samples in total.

3.2 Model and Activations

We use BERT-base-uncased (Devlin et al., 2019) with all weights frozen throughout. For each input text we extract the CLS-token hidden state at every transformer layer $l \in \{1, \ldots, 12\}$, giving a per-layer vector $\mathbf{h}^{(l)} \in \mathbb{R}^{768}$. The full feature representation is the concatenation

$$\mathbf{z} = \left[\mathbf{h}^{(1)}; \ldots; \mathbf{h}^{(12)}\right] \in \mathbb{R}^{9,216}.$$

We refer to individual dimensions of $\mathbf{z}$ as *neurons*; neuron index i resides in layer $\lfloor i/768 \rfloor + 1$ and position $i \bmod 768$ within that layer. Using the frozen pretrained model (rather than a fine-tuned detector) ensures that any informative features we identify are present in general-purpose BERT representations, not artefacts of task-specific fine-tuning.

3.3 Protocol: L1 Selection, L2 Evaluation

Following Gurnee et al. (2023), for each (seed, fold) cell we fit two independent probes on the training split. (1) *L1 selection probe*: L1-regularised logistic regression ($C = 0.005$, liblinear solver) on all 9,216 features. Neurons with non-zero coefficients form the *candidate set* $\mathcal{S}$; neurons appearing in $\mathcal{S}$ in at least 50% of the 15 cells are *stable neurons*. (2) *L2 evaluation probe*: L2-regularised logistic regression (lbfgs) on the full $\mathbf{z}$. This probe is held frozen for all downstream experiments to ensure that intervention effects are measured against a fixed decision function.

We use 5-fold stratified cross-validation stratified by (label $\times$ domain) with seeds $\{42, 123, 456\}$, yielding 15 cells per (generator, experiment). All reported metrics are 15-cell means $\pm$ standard deviation unless otherwise noted.

3.4 Stability-Driven Hyperparameter Selection

The regularisation strength $C = 0.005$ and sample size $N = 7{,}500$ were selected via a multi-generator stability sweep conducted prior to any results analysis, following the criterion of Gurnee et al. (2023). We drew $K = 15$ stratified subsamples from each of three generators (`gpt4`, `llama-chat`, `mistral-chat`), computed mean pairwise Jaccard similarity between selection sets across subsamples, and chose the (C, N) pair that maximised the minimum Jaccard across generators. At $C = 0.005$, $N = 7{,}500$, the minimum-over-generators Jaccard is 0.724 — the highest achieved by any C at every N tested. The full sweep is reported in Appendix B.

4 Sparse Detection Circuits in Frozen BERT

Table 1 reports sparse-probe results across all six generators. Between 45 and 62 neurons — 0.5–0.9% of the 9,216-feature CLS representation — are stable across all 15 cells, yet the L2 evaluation probe achieves 90.5–98.8% validation accuracy (AUC-ROC 0.968–0.999). Selection is highly reproducible: pairwise fold Jaccard ranges 0.67–0.76 across 105 fold-pairs per generator, compared to a random-null expectation of ∼0.003–0.005 for sets of this size drawn from 9,216 features, placing within-generator stability at 130–200× above chance. The selection pool is narrow: between 91 neurons ever selected (llama-chat) and 146 (gpt2), indicating that the probe consistently returns to a small region of the feature space.

Detection difficulty varies systematically across generators. GPT-4 is easiest to detect (98.8% val acc, AUC 0.999), followed by LLaMA-2-Chat (98.6%) and Mistral-7B-Instruct (97.1%); all three are instruction-tuned and produce outputs that diverge markedly from human writing in BERT's frozen representation space. MPT-7B (97.2%) is close behind despite being a pure-base generator, with 52 stable neurons forming a compact and highly reproducible selection set. GPT-2 sits at the mid-point (95.9%) with the loosest fold Jaccard (0.670) and largest observed pool (146 neurons) among all six generators, suggesting a noisier but stable feature landscape for this base model. Cohere-Command-Chat is the structural difficulty floor at 90.5% val accuracy and AUC 0.968, with the highest standard deviation on n_{sel} (CV 4.3%),

Generator	Val acc	n_{sel}	Jaccard	n_{stable} / n_{obs}
gpt4	0.988 ± 0.003	74.1 ± 2.9	0.763 ± 0.037	60 / 106
gpt2	0.959 ± 0.003	84.7 ± 4.1	0.670 ± 0.042	60 / 146
mpt	0.972 ± 0.003	61.3 ± 3.1	0.753 ± 0.040	52 / 93
mistral-chat	0.971 ± 0.004	57.8 ± 2.9	0.724 ± 0.046	45 / 99
llama-chat	0.986 ± 0.003	63.6 ± 1.7	0.756 ± 0.040	48 / 91
cohere-chat	0.905 ± 0.008	72.3 ± 3.1	0.749 ± 0.039	62 / 113

Table 1: Sparse-probe results across six generators: 15-cell means ± std (5 folds × 3 seeds). Val acc is the L2 evaluation probe accuracy (all 9,216 features). n_{sel}: mean candidate set size per cell. Jaccard: pairwise fold Jaccard over 105 fold-pairs (mpt estimated analytically from per-neuron stability frequencies). n_{stable}: neurons in ≥50% of cells; n_{obs}: total distinct neurons selected at least once.

consistent with its smaller per-axis signal magnitude documented in Section 7.

5 Causal Sufficiency via Activation Patching

This section tests whether the selected neurons are *causally sufficient* to produce AI-text predictions. If transplanting their activations from a real AI sample into a human sample causes the evaluation probe to flip its prediction, the selected set carries the detection signal rather than merely correlating with it. This is a strictly stronger claim than probe accuracy alone.

5.1 Patching Protocol

For each (seed, fold) cell we identify *valid human* samples: test-fold human texts that share a domain with at least one test-fold AI sample. For each of $n_{\text{shuffles}} = 20$ random within-domain donor permutations we construct a patched activation $\tilde{\mathbf{z}}$ by replacing the selected neurons with those from the AI donor:

$$\tilde{\mathbf{z}}_i = \begin{cases} \mathbf{z}_i^{\text{donor}} & i \in \mathcal{S} \\ \mathbf{z}_i^{\text{human}} & i \notin \mathcal{S} \end{cases}$$

A *flip* occurs when the frozen evaluation probe classifies $\tilde{\mathbf{z}}$ as AI but the unpatched $\mathbf{z}^{\text{human}}$ as human. The flip rate is the fraction of valid human samples that flip, averaged over the 20 donor permutations and then over the 15 cells. Donor sampling is restricted to the same domain as the target, following the on-manifold recommendation of Heimersheim and Nanda (2024) to avoid injecting out-of-distribution activations.

The *random-k baseline* draws 20 random subsets of size k from the 9,216 features and runs the

k	gpt4	gpt2	mpt	mistral	llama	cohere
1	0.02	0.05	0.10	0.06	0.07	0.13
5	0.09	0.17	0.25	0.20	0.14	0.58
10	0.15	0.29	0.43	0.34	0.27	0.99
20	0.35	0.58	0.79	0.56	0.49	2.02
50	0.80	1.71	1.52	1.48	0.86	5.43
full	$1.23_{\pm 0.34}$	$3.13_{\pm 0.56}$	$1.59_{\pm 0.23}$	$1.73_{\pm 0.28}$	$1.07_{\pm 0.38}$	$8.15_{\pm 1.32}$

Table 2: Selected flip rate (%) across the k-sweep, 15-cell means (5 folds $\times$ 3 seeds). k = full equals each cell's actual $|\mathcal{S}|$ (53–92 neurons); std shown only for k = full. mpt $k <$ full values from a representative fold. Flip rate is monotonically increasing in k for every generator.

identical procedure, providing a null that controls for the number of patched dimensions. We sweep $k \in \{1, 5, 10, 20, 50, \text{full}\}$ where k = full equals the cell's actual $|\mathcal{S}|$ (53–92 neurons depending on generator and cell).

5.2 Flip-Rate Results

Table 2 reports selected flip rates across the k-sweep. The result is consistent across all six generators: flip rate is monotonically increasing in k, and the selected set outperforms the random-k baseline at every k value. At k = full, patching the selected neurons causes 1.07–8.15% of human samples to be reclassified as AI-generated. Against a random baseline of 0.08–0.78% at the same k, this yields a ratio of 10–23$\times$ on five of six generators. For gpt4 the ratio is $\sim$125$\times$ but with standard deviation $\approx$265, because the random-k flip rate is near zero ($\sim 8 \times 10^{-4}$) and small denominator fluctuations dominate; the selected flip rate of 1.23% is the robust signal there. MPT-7B falls at the lower end of the range (1.59%, $\sim$10$\times$ above random), consistent with its moderate detection difficulty.

Cohere-Command-Chat is the most patchable generator (8.15% at k = full, $\pm$1.32%), consistent with its lowest baseline accuracy (90.5%): a probe operating nearer its decision boundary crosses it under a smaller perturbation. LLaMA-2-Chat is the hardest to flip (1.07%, $\pm$0.38%) despite its high baseline accuracy (98.6%), suggesting its detection signal is more broadly distributed. Flip rate scales approximately log-linearly with k across all generators, indicating that each added neuron contributes incrementally rather than the effect concentrating in a few critical units.

5.3 Robustness Checks

[TODO: Run cross-domain donor baseline before submission: replace same-domain donors with random cross-domain AI donors and measure flip rate gap vs same-domain. Expected to confirm the effect is not driven by domain-level activation shifts. Also consider reporting zero-ablation alongside mean-ablation in §6 per Heimersheim and Nanda (2024) off-manifold concern.]

6 Necessity via Mean Ablation

To complement the sufficiency test we ask the converse question: is the selected set *necessary*? We answer this via mean ablation, replacing the selected neurons in the test-fold activations with their training-fold mean (computed on the training split to avoid leakage) and re-evaluating the frozen L2 probe. Table 3 reports accuracy drop at k = full alongside the random-k baseline.

For four of five generators the result is negative. At k = full, ablating the selected set causes accuracy to drop by only 0.03–0.11 pp — within noise of the 0.00–0.06 pp random-k baseline. The selected neurons are not individually necessary: the L2 evaluation probe, which uses all 9,216 features, retains thousands of redundant correlates that compensate for the removed subset.

Cohere-Command-Chat is the exception. Its full-set ablation drop is 1.06 ± 0.54 pp — roughly 18$\times$ the random baseline of 0.06 pp — and the drop grows monotonically from $k = 5$ upward, unlike the other four generators where the curve is flat. This pattern is consistent with the other cohere signals (lowest baseline accuracy, smallest mean-difference norm, largest probe weight norm) and suggests its detection representation is *partially* localised rather than fully distributed.

There is no contradiction between the small ablation drops here and the large patching flip rates in Section 5. Ablation tests necessity *against a probe that has access to all other features*: removing $\leq$1% of 9,216 features leaves thousands of correlated dimensions intact. Patching tests sufficiency by *injecting wrong-class signal that competes with the remainder of the network*. A representation that is redundantly distributed is exactly what would produce small ablation drops alongside meaningful patching effects. We interpret the joint result as: the selected neurons carry genuine AI-detection signal, but the same signal is also distributed across the broader feature space.

Generator	Baseline acc	Selected drop (k=full)	Random drop (k=full)
gpt4	0.988	$+0.03 \pm 0.12$ pp	$+0.01 \pm 0.06$ pp
gpt2	0.959	$+0.11 \pm 0.31$ pp	$+0.00 \pm 0.00$ pp
mpt	0.972	$+0.05 \pm 0.11$ pp	$+0.00 \pm 0.02$ pp
mistral-chat	0.971	$+0.10 \pm 0.21$ pp	$+0.06 \pm 0.10$ pp
llama-chat	0.986	$+0.11 \pm 0.14$ pp	$+0.00 \pm 0.00$ pp
cohere-chat	0.905	$\mathbf{+1.06 \pm 0.54}$ pp	$+0.02 \pm 0.04$ pp

Table 3: Mean-ablation accuracy drop at k = full, 15-cell means $\pm$ std. Drop = baseline $-$ ablated accuracy (positive = accuracy decreases). Random drop is the mean over 20 random k-subsets of the same size. Cohere-chat is the only generator with a drop substantially above the random baseline.

Layer	gpt4	gpt2	mpt	mistral	llama	cohere
1–10	36	41	**44**	24	30	40
11	6	**11**	6	5	2	3
12	**18**	8	2	**16**	**16**	**19**
% in L12	30.0	13.3	3.8	35.6	33.3	30.6

Table 4: Stable-neuron counts by layer group across six generators. Bold indicates the peak layer group for each generator. Instruction-tuned generators (gpt4, mistral, llama, cohere) concentrate 30–36% of stable neurons in layer 12; both pure-base generators (gpt2, mpt) have $\leq$14% in layer 12.

A methodological note: mean ablation replaces activations with out-of-distribution values that fall off the natural data manifold, which can introduce artefacts (Heimersheim and Nanda, 2024). We treat the necessity result as conservative and plan to report zero-ablation as a robustness check (Section 5.3).

7 Cross-Generator Circuit Geometry

7.1 Layer Distribution of Stable Neurons

Table 4 shows the layer-by-layer distribution of stable neurons across all six generators. A clear asymmetry emerges between the four instruction-tuned generators and the two pure-base generators. For gpt4, mistral-chat, llama-chat, and cohere-chat, the single most populated layer is layer 12, which accounts for 30–36% of each generator's stable set (16–19 neurons). Both pure-base generators are exceptions: GPT-2 peaks at layer 11 with 11 stable neurons (18.3% of its stable set) and has only 13.3% in layer 12; MPT-7B has just 2 stable neurons in layer 12 (3.8%) and distributes its signal more broadly across earlier layers (layers 1, 8, and 9 dominate).

Across both pure-base generators, the shared pattern is markedly low layer-12 concentration (3.8–13.3%) compared to any instruction-tuned generator (30–36%). This 2-vs-4 contrast suggests that BERT's final layer functions as a generator-specific classification register preferentially activated by instruction-tuned outputs, while base-model outputs engage BERT's earlier and mid-depth representations more strongly. The specific peak layer differs between gpt2 (L11) and mpt (L1/L8), indicating the effect is not a rigid "base generators peak at L11" rule but rather a consistent *avoidance* of L12 by non-instruction-tuned generators.

7.2 Stable-Set Overlap Across Generators

Table 5 reports pairwise Jaccard similarity between stable sets across all fifteen generator pairs. In absolute terms, overlap is low: Jaccard ranges from 0.00 to 0.24, consistent with generator-specific circuits. However, the appropriate baseline is not zero but the expected Jaccard for two random sets of the same sizes drawn from 9,216 features, which is approximately 0.003 per pair. Relative to this null, the mean observed Jaccard of 0.073 is $\sim$24$\times$ above chance, indicating a partial shared substrate despite the generator-specific structure.

The data reveal a clear bipartite structure. Among instruction-tuned generators, pairwise Jaccard ranges 0.08–0.24 (27–86$\times$ chance), with the strongest pairs being cohere–mistral (86$\times$) and llama–mistral (64$\times$). Among the two pure-base generators, mpt and gpt2 share 12 neurons (J = 0.120, 40$\times$ chance) — higher than several instruction–instruction pairs — confirming that the base models form a coherent sub-cluster. In contrast, every base-vs-instruction-tuned pair has near-zero or zero overlap: mpt has J = 0.000 with both gpt4 and cohere-chat, and J $\leq$ 0.011 with mistral and llama. Three pairs now have zero overlap (mpt–gpt4, mpt–cohere, gpt2–gpt4), all involving at least one base generator. This base-vs-instruction-tuned partition explains the bulk of the cross-generator circuit geometry.

7.3 CAV Diagnostics

To characterise the geometry of the detection signal within each stable set we compute a concept activation vector (CAV) as the mean difference between AI and human activations restricted to the stable neurons, and compare it to the L1 probe weight vector. Table 6 reports cosine similarity between the CAV and the probe weight, logistic-regression accuracy on the CAV projection, and the norms of

Pair	Jaccard	× chance
cohere – mistral	0.244	86×
llama – mistral	0.163	64×
gpt4 – mistral	0.141	51×
gpt4 – llama	0.113	39×
mpt – gpt2	**0.120**	**40×**
cohere – gpt4	0.109	33×
cohere – llama	0.078	27×
gpt2 – mistral	0.050	18×
cohere – gpt2	0.034	10×
gpt2 – llama	0.019	7×
mpt – mistral	0.010	4×
mpt – llama	0.010	4×
gpt2 – gpt4	0.000	0×
mpt – gpt4	0.000	0×
mpt – cohere	0.000	0×
Mean	0.073	∼24×

Table 5: Pairwise Jaccard similarity between stable sets (15 generator pairs). × chance is observed Jaccard divided by the analytic random-null Jaccard for two sets of the same sizes drawn from 9,216 features (≈0.003 per pair). The mpt–gpt2 pair (bold) is the highest base×base overlap; all base×instruction-tuned pairs cluster at the bottom.

Generator	Cosine (CAV, w)	LR acc	$\|\Delta\mu\|$	$\|w\|$
gpt4	0.140	0.985	7.34	47.9
gpt2	0.071	0.969	6.84	69.1
mpt	0.065	0.971	**9.52**	**53.9**
mistral-chat	0.111	0.963	6.38	58.8
llama-chat	0.129	0.990	6.57	47.2
cohere-chat	**0.057**	**0.903**	4.69	107.0

Table 6: CAV diagnostics on per-generator stable sets. Cosine: similarity between the mean-difference CAV and the L1 probe weight vector. LR acc: logistic-regression accuracy using only the CAV projection. $\|\Delta\mu\|$: norm of the AI−human mean-difference vector (bold = largest). $\|w\|$: L1 probe weight norm (bold = smallest). Single-fit per generator; no standard errors available.

both vectors.

Cohere-chat has the lowest CAV cosine similarity (0.057), the lowest CAV-LR accuracy (0.903), and by far the largest probe weight norm (107), consistent with a diffuse signal compensated by large weights. MPT-7B presents an interesting contrast: it has the lowest cosine (0.065, near cohere's) but the *largest* mean-difference norm (9.52) and smallest weight norm (53.9) among all six generators — suggesting that mpt's detection signal is strong in per-axis magnitude but geometrically misaligned with the probe direction relative to instruction-tuned generators. GPT-2 also shows a low CAV cosine (0.071) and elevated weight norm (69), placing both base generators at the low-cosine end. Instruction-tuned generators (gpt4, llama-chat, mistral-chat) cluster at higher cosine (0.11–0.14) with moderate weight norms (47–59). CAV diagnostics are currently single-fit per generator and carry no standard errors; wrapping them in the multi-seed loop is a planned runner enhancement.

8 Discussion

Redundancy and sufficiency are not in tension. The joint pattern across Sections 5 and 6 — large patching flip rates alongside negligible ablation drops on 4/5 generators — has a natural reading: AI-detection signal is *redundantly distributed*. The selected neurons carry genuine causal signal (evidenced by patching), but the same information is backed up across the remaining 9,000+ features in the residual stream (evidenced by the small ablation drop). Ablation understates the role of the selected set because removing ≤1% of a 9,216-dimensional representation leaves thousands of correlated features intact. Patching reveals that role precisely because it injects wrong-class signal that must compete with the rest of the network — a harder test. The phrase "easy to inject, hard to erase" captures this geometry: the selected neurons are sufficient but not necessary.

Outliers as structural evidence. Cohere-Command-Chat deviates from other generators on multiple axes: weakest per-axis signal ($\|\Delta\mu\| = 4.69$, CAV cosine 0.057), largest probe weights (norm 107), most patchable output (8.15% flip rate), and the only generator with a substantial ablation cost (1.06 pp). Together these point to a signal encoded diffusely with small per-feature magnitude — the probe compensates via large weights, patching is easy because the decision boundary is near, and ablation matters more because fewer redundant correlates exist. The two pure-base generators, GPT-2 and MPT-7B, form a coherent structural cluster: both have near-zero or zero stable-set overlap with instruction-tuned generators (J ≤ 0.05), yet they share 12 neurons with each other (J = 0.120, 40× chance); both avoid layer 12 (3.8% and 13.3% respectively, versus 30–36% for instruction-tuned models); and both exhibit the two largest restricted-probe gaps of the six generators (mpt 4.68 pp, gpt2 3.63 pp, versus ≤1.1 pp for the others). This consistency across two independent base models strengthens the interpretation that post-training

alignment leaves a layer-12 footprint that is absent in base-only decoding.

Relation to confounder-suppression approaches. Borile and Abrate (2025) identify and suppress neurons correlated with domain or style artefacts in a fine-tuned BERT encoder to improve out-of-distribution detection. Our work is complementary along the intervention axis: their question is "which neurons should we remove to make the detector more robust?", while ours is "which neurons does the pretrained model already use, and are they causally sufficient?" The two approaches converge on the same empirical substrate — a small subset of BERT neurons carries disproportionate detection-relevant information — but reach it from opposite directions, necessity-for-robustness versus sufficiency-for-detection.

9 Limitations

Single encoder architecture. All experiments use frozen BERT-base-uncased. Whether the stable-neuron structure and patching effects extend to larger encoders (RoBERTa-large, DeBERTa-v3) or encoder-decoder architectures is an open question. The frozen setting is a deliberate design choice — it isolates what the pretrained representation already encodes — but it means results cannot be directly compared to fine-tuned detectors without controlling for this variable.

Mean ablation is off-manifold. Replacing selected neurons with their training-mean pushes activations off the natural data manifold, which can introduce artefacts unrelated to the detection signal (Heimersheim and Nanda, 2024). We treat the necessity results as conservative lower bounds and plan to report zero-ablation as a robustness check (Section 5.3); a full resample-ablation comparison remains future work.

Layer-asymmetry claim across two base generators. The observation that instruction-tuned generators concentrate 30–36% of stable neurons in layer 12 while pure-base generators have $\leq$14% is now supported by two independent base models (GPT-2 and MPT-7B). However, the two base generators differ in their peak layer (GPT-2: L11; MPT-7B: L1/L8), so the claim is best stated as L12-avoidance rather than a strict L11-peaking rule. We frame this as a consistent L12 footprint of post-training alignment rather than a mechanistically precise claim about which specific earlier layer is activated.

Adversarial robustness not evaluated. RAID includes eleven adversarial rephrasing attacks (paraphrase, homoglyph, synonym substitution, etc.) that substantially reduce detector accuracy (Dugan et al., 2024). Whether the stable neurons identified here remain causal under these attacks is left to future work; the present analysis covers only the non-adversarial RAID subset.

Scope of causal claim. Activation patching demonstrates that the selected neurons are sufficient to shift probe predictions, but does not constitute a full circuit-level proof of causality in the sense of Meng et al. (2022). The patching intervention is at the level of individual neuron activations, not a complete mechanistic decomposition of the computation. We make no claims about what linguistic properties these neurons encode or why they are informative.

Ethics Statement

Acknowledgments

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A Pipeline Details

B Stability Sweep

C Per-Cell Metrics

D AUC vs L1 Selection

E Restricted Probe (Smaller-Model vs Ablate-Complement)

F Per-Domain Patching Breakdown